# Week 12: Assignment 12

**Topic:** Seq2Seq, Attention & Transformers **Status:** Submitted & Verified (Score: 20/20)

## Case Study 1: Clinic Discharge Note Conversion (Questions 1 to 5)

A regional hospital converts doctor discharge notes into patient-facing versions using an encoder-decoder architecture. Early prototypes passed only one final summary vector, causing long notes to lose early warnings like "avoid aspirin."

**1. In the early prototype, what is the main risk of using only one final summary vector?**

- The decoder must output every source word unchanged.
- The encoder is unable to read numerical tokens.
- ✔ **Important source details may be compressed away.**
- The vocabulary becomes empty during generation.

**2. What does the encoder mainly provide to the decoder in this sequence model?**

- A sorted list of output tokens for the note
- ✔ **Representations derived from the source note**
- The final patient label after thresholding
- A replacement for the training loss function

**3. Why is the decoder described as producing the converted note token by token?**

- It ignores all tokens generated earlier.
- ✔ **Each new token can depend on earlier output tokens.**
- It computes the encoder states after every output only.
- Every output token must have the same probability.

**4. Which change most directly helps the decoder recover a warning sentence from early in a long note?**

- Removing all source-token representations
- Replacing the decoder with a lookup table
- Using only the last generated token as input
- ✔ **Allowing access to relevant encoder states**

**5. If a long converted note omits an early allergy warning, what does that suggest about the fixed-summary design?**

- ✔ **The final summary did not preserve all needed information.**
- The output language has no word order requirement at any step.
- The source note was processed without any token representations.
- The decoder cannot produce medical vocabulary for short notes.

### Case Study 2: Maintenance Ticket Digest (Questions 6 to 10)

A support desk generates digests from tickets. Softmax alignment scores assign attention weights (0.10, 0.25, 0.15, 0.50) across four ticket sections to compute the context vector.

**6. What is the role of the attention weights in this decoder step?**

- ○ They replace all encoder states with fixed zero vectors.
- ○ They set the number of output classes for the digest.
- ✔ **They specify how much each section contributes.**
- ○ They sort the ticket sections into alphabetical order.

**7. Why is softmax applied to the alignment scores?**

- ✔ **To convert scores into normalized nonnegative weights.**
- ○ To remove the decoder hidden state from every step.
- ○ To force the largest raw score to become exactly one.
- ○ To make all ticket sections contribute with equal weights.

**8. Which ticket section contributes most to this output token? (Weights: 0.10, 0.25, 0.15, 0.50)**

- ○ Section 1
- ○ Section 2
- ○ Section 3
- ✔ **Section 4**

**9. How is the context vector computed after these weights are available?**

- ○ By choosing the first encoder state only
- ✔ **By forming a weighted sum of encoder states**
- ○ By averaging decoder outputs from all steps
- ○ By discarding the highest-weight section

**10. If the support team expects the replacement-part section to drive this token, what attention pattern would support that interpretation?**

- ○ All four sections receiving negative attention weights
- ○ The decoder state being removed from the scoring step
- ✔ **The replacement-part section receiving the largest weight**
- ○ Every ticket section receiving exactly the same weight

### Case Study 3: Procurement Rule Checking (Questions 11 to 15)

A tender-review model uses self-attention with Query, Key, and Value vectors. Attention scores use scaled dot-products and softmax.

**11. What advantage does self-attention give for checking references across distant clauses?**

- It forces every token to use only its previous neighbor.
- It removes all long-distance comparisons.
- It replaces the document with a single class label.

✔ **It lets tokens compare directly with other tokens.**

**12. In the query-key comparison, what do the resulting scores represent?**

- The final number of tender documents reviewed

✔ **Relevance between one token and another token**

- The optimizer's learning-rate value during training
- The length of the output vocabulary for labels

**13. What is the purpose of scaling the dot product by square root of d_k?**

✔ **To keep score magnitudes controlled before softmax**

- To erase the effect of all value vectors in the layer
- To make positional encodings unnecessary for ordering
- To force all attention scores to become identical

**14. What role do Value vectors play after attention weights are computed?**

- They choose the batch size for training runs.
- They remove token-order information from clauses.

✔ **They provide the content that gets combined.**

- They store only the document identifier value.

**15. Why are positional encodings needed in this tender-review model?**

- They delete repeated words from the document.
- They replace softmax in attention scoring.
- They make every token attend to itself only.

✔ **They supply information about token order.**

### Case Study 4: Campus Feedback Review (Questions 16 to 20)

A university analyzes student feedback reports using two-level attention (word-level and sentence-level) and multiple attention heads.

**16. Why is a two-level representation natural for these reports?**

- Reports contain only a single unordered word list.
- ✔ **Words form sentences, and sentences form reports.**
- The model must ignore sentence boundaries.
- Every paragraph has exactly one token.

**17. What does word-level attention help identify?**

- The number of reports in the dataset split
- The final optimizer update rule for training
- The batch size used for report classification
- ✔ **Important words or phrases inside a sentence**

**18. What does sentence-level attention help identify?**

- ✔ **Sentences that are most useful for the report decision**
- Characters that should be removed from each word
- Output tokens generated by a decoder layer
- The number of attention heads in the layer

**19. Why can multiple attention heads be useful in this feedback model?**

- They force all heads to learn identical attention weights.
- They delete word-level and sentence-level evidence.
- ✔ **They allow different evidence patterns to be tracked.**
- They make positional information impossible to use.

**20. If sentence attention is almost uniform across a long report, what concern should the team consider?**

- The report contains no sentence representations.
- ✔ **The model may not be selecting the most informative sentences.**
- Multi-head attention cannot operate on textual inputs.
- Word-level attention must always be zero for all words.